

Introduction to Robot Autonomy

By Jonathan Diller

Course Topics

- Robotics Operating System (ROS2)[†]
- Perception
- Localizing
- Path Planning
- Control
- Advanced Topics:
 - Mapping^{*}
 - Model Predictive Control^{*}
 - Mission Planning^{*}

^{*} Topic does not include Build-Your-Own

[†] Topic does not include Employ-Existing-Tools

Week-to-Week Format

First Week

Build-Your-Own!

Monday:

- Motivate new topic
- Formal lecture
- ✓ In-class worksheet

Wednesday:

- Reinforce lecture
- ✓ Concepts Quiz
- In-class example led by instructor

Friday:

- Introduction to lab
- ✓ Students implement their own version of the system¹

Second Week

Employ-Existing-Tools

Monday:

- Discussion on advanced topics
- Introduction to advanced tools
- ✓ 60-second essay on advanced topic

Wednesday:

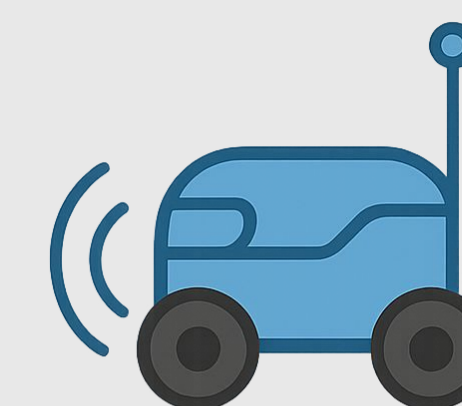
- Introduction to lab
- ✓ Use existing implementation of system for lab project^{1,2}

Friday:

- ✓ Use existing implementation of system for lab project
- Students demonstrate final implementation to instructor/TAs

Course Learning Outcomes (CLOs):

- Students will create their own ROS packages that make up the main elements of a robot autonomy stack.
- Students will assemble a vision module that can find transformations between two laser scans for a robot with a LIDAR using the Iterative Closest Point algorithm.
- Students will develop a particle filter for localizing a robot in a known environment given a previous robot state, odometry measurements, and a transposition from visual data.
- Students will develop a path planning algorithm of their choice for holonomic robots that finds a collision-free path from a start location to a goal location.
- Students will design a PID controller for a holonomic robot to develop an ego-planner that follows a predetermined trajectory.
- Students will construct a complete autonomy stack by combining their lab assignments on Perception, Localization, Control, and Planning and demonstrating how these parts function together to control a robot.



PERCEPTION

LOCALIZATION

PLANNING

CONTROL

References

1. Michael J. Hannafin and Susan M. Land. "Technology and student-centered learning in higher education: Issues and practices." Journal of Computing in Higher Education, 2000.
2. David S. Touretzky "Robotics for computer scientists: what's the big idea?" Computer Science Education, 2013.